

When Is The Trend Your Friend? Understanding Momentum Across Macro Markets

Special Report

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Dear Client,

BCA's Global Investment Strategy (GIS) team believes in full transparency and accountability. With that in mind, we will be launching our GIS FullView Macro Portfolio next week. The portfolio combines our own subjective views with the technical indicators discussed in today's report. It will also incorporate the profit or loss from all our tactical macro trade recommendations. We will track it against a standard 60/40 portfolio so that you can see whether our views and trades are adding value or not.

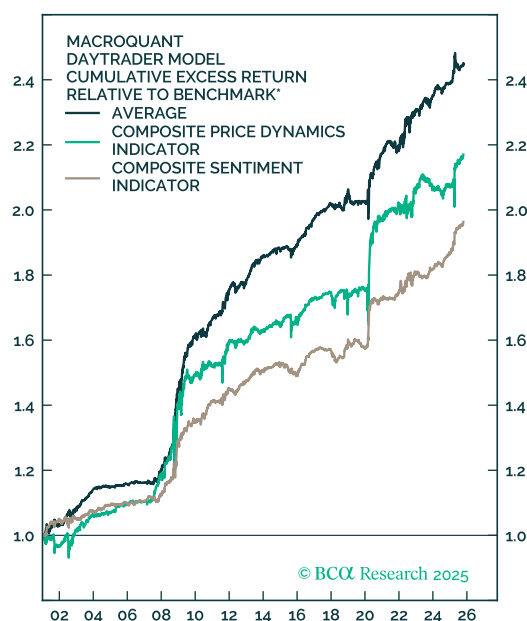
Best regards,
Peter Berezin
Chief Global Strategist

Executive Summary

Top Takeaway: Mean-reversion strategies work best for equities and bonds, while trend-following strategies work best for commodities and, to a lesser extent, currencies.

- We examine how a variety of momentum strategies perform for different assets using moving average rules ranging from 1 day to 200 days.
- For the S&P 500, mean-reversion (i.e., "buy on weakness") strategies work well over short-term horizons.
- Treasury yields are subject to short-term mean reversion. A strategy of buying TLT and shorting SPY whenever the former underperformed the latter over the prior few days has been quite profitable.
- A trend-following (i.e., "buy on strength") strategy works reasonably well for the US dollar and even better for oil and precious metals. Perhaps not surprisingly, trend-following strategies also work well for crypto.

Combining Signals From Momentum And Sentiment Data Amplifies Model Performance



* SHOWN INDEXED TO JAN 4, 2001 = 1.0.
BENCHMARK DEFINED AS A STRATEGY THAT IS ON AVERAGE AS BULLISH AS THE MACROQUANT DAYTRADER INDICATOR (50% EQUITY HOLDING FOR AVERAGE, 51% EQUITY HOLDING FOR COMPOSITE PRICE DYNAMICS INDICATOR, AND 44% EQUITY HOLDING FOR COMPOSITE SENTIMENT INDICATOR).
SOURCE: MACROQUANT MULTIVERSE EDITION (DAYTRADER).

When Is The Trend Your Friend? Understanding Momentum Across Macro Markets

Macro Momentum: A Primer

The efficacy of return-continuation (also known as “trend-following” or “buy on strength”) strategies versus mean-reversion (also known as “buy on weakness”) strategies depends on the asset under consideration, the duration of the trend in question, and the broader state of the macro economy (i.e., whether one is in a recession or not).

In this report, we lay out the key factors that investors need to know. We conclude by highlighting which assets our indicators are currently most bullish and most bearish on.

To conduct our analysis, we look at the daily return for each asset relative to its 1-day moving average (i.e., prior day’s price), as well as its 10-day, 50-day, and 200-day moving averages. To allow for easy comparisons across assets, we calculate a z-score for each price deviation. The higher the z-score, the higher is the price relative to its moving average.

If either the z-score is positive and the subsequent day’s return is above average, or the z-score is negative and the subsequent day’s return is below average, then the asset displays return-continuation (i.e., a trend-following strategy works best). Conversely, if either the z-score is negative and the subsequent day’s return is above average, or the z-score is positive and the subsequent day’s return is below average, then the asset displays mean-reversion (a buy-on-weakness strategy works best).

In addition to positive and negative z-scores, we also consider the average net return for cutoffs where the z-score is below -0.5 and above 0.5; below -1 and above 1, below -1.5 and above 1.5, and below -2 and above 2. We conduct our analysis for the complete sample (2001 to the present) and a more recent sample (2020 to the present). The **Appendix** at the back of the report shows the key results, which we discuss below.

THE BIG PICTURE One Week to Go!

REGISTER

S&P 500: Mean-Reversion Dominates

Looking at the S&P 500 since 2001, the index gained 10 bps on days when it fell during the previous day but lost 2 bps on days when it rose during the previous day (for a net return of 12 bps). Since 2020, the net return has averaged 13 bps.

The net return rises if we look at sharper cutoffs. For example, if we limit the analysis only to cases where the z-score for the prior day was either below -0.5 or above 0.5, we see that since 2001, the index gained 15 bps on days when it fell more than 0.5 standard deviations the previous day, but lost 5 bps on days when it rose more than 0.5 standard deviations the previous day (for a net return of 20 bps). The net return rises to 35 bps for z-scores below -1 and above 1, 50 bps for z-scores below -1.5 and above 1.5, and 83 bps for z-scores below -2 and above 2. The effects taper off if one looks at longer moving averages but the mean-reversion strategy still generates meaningful net returns.

Perhaps surprisingly, a buy-on-weakness strategy works better in equity bear markets than in bull markets. This is because volatility tends to be higher during bear markets, allowing for more situations where stocks fall a lot on one day, only to temporarily snap back the following day. Even during the extreme bear market from October 9, 2007, to March 9, 2009 – a period when the S&P fell 58% – stocks rose 49% of the days.

Momentum at the Equity Sector and Region Levels

Our analysis did not generate reliable results for the performance of US equity sectors relative to the S&P 500 index based on daily data. Taking a somewhat longer-term view by focusing on monthly data, return continuation appears to be more common than mean reversion. This is especially true for highly defensive sectors such as consumer staples as well as very cyclical ones such as materials (**Table 1**).

TABLE 1

The Efficacy Of Trend-Following Strategies For US Equity Sectors Over Longer-Time Horizons

	Net Forward 1-Month Return To A Trend-Following Strategy Based On Momentum*			
	Prior 1-Month Relative Return	Prior 3-Month Relative Return	Prior 6-Month Relative Return	Prior 12-Month Relative Return
Communication Services/S&P 500 (XLC/SPY)	-0.12	0.86	0.80	0.91
Consumer Discretionary/S&P 500 (XLY/SPY)	0.23	0.27	-0.12	0.13
Consumer Staples/S&P 500 (XLP/SPY)	0.53	0.16	0.45	0.41
Energy/S&P 500 (XLE/SPY)	0.28	0.14	0.06	1.02
Financials/S&P 500 (XLF/SPY)	-0.16	0.37	0.14	0.26
Health Care/S&P 500 (XLV/SPY)	0.35	-0.08	0.04	0.14
Industrials/S&P 500 (XLI/SPY)	-0.07	-0.04	-0.17	-0.35
IT/S&P 500 (XLK/SPY)	-0.69	0.43	0.53	0.20
Materials/S&P 500 (XLB/SPY)	0.31	0.54	0.34	0.12
Real Estate/S&P 500 (XLRE/SPY)	-0.07	-0.46	0.30	0.48
Utilities/S&P 500 (XLU/SPY)	-0.42	-0.18	-0.02	-0.01

* BASED ON DATA SINCE SEPTEMBER 2018 FOR COMMUNICATION SERVICES, NOVEMBER 2001 FOR REAL ESTATE, AND 1995 FOR ALL REMAINING SECTORS. NOTE: A POSITIVE (GREEN) NET AVERAGE FORWARD RETURN MEANS THAT THE SECTOR OUTPERFORMED THE BROAD MARKET WHEN THE RELATIVE PRICE MOMENTUM WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A TREND-FOLLOWING STRATEGY. IN CONTRAST, A NEGATIVE (RED) NET AVERAGE FORWARD RETURN MEANS THAT THE SECTOR UNDERPERFORMED THE BROAD MARKET WHEN THE RELATIVE PRICE MOMENTUM WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A MEAN-REVERSION STRATEGY. SOURCE: MACROQUANT (SECTOR SELECTOR).

TABLE 2

The Efficacy Of Trend-Following Strategies For Equity Regions Over Longer-Time Horizons

	Net Forward 1-Month Return To A Trend-Following Strategy Based on Momentum*				
	Prior 1-Month Relative Return	Prior 3-Month Relative Return	Prior 6-Month Relative Return	Prior 12-Month Relative Return	Prior 18-Month Relative Return
US/Global (SPY/ACWI)	0.23	0.34	0.32	0.34	0.33
Euro Area/Global (EZU/ACWI)	0.42	0.12	0.12	-0.06	0.03
UK/Global (EWU/ACWI)	-0.05	-0.14	0.02	0.15	0.18
Canada/Global (EWC/ACWI)	0.00	0.30	0.32	0.38	0.37
Australia/Global (EWA/ACWI)	-0.40	-0.32	0.24	0.27	0.28
Japan/Global (EWJ/ACWI)	-0.13	0.11	0.06	-0.72	-0.83
EM/Global (EEM/ACWI)	0.84	0.72	0.75	0.93	0.45

* BASED ON DATA SINCE 1995.

NOTE: A POSITIVE (GREEN) NET AVERAGE FORWARD RETURN MEANS THAT THE REGION OUTPERFORMED GLOBAL EQUITIES WHEN THE RELATIVE PRICE MOMENTUM WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A TREND-FOLLOWING STRATEGY. IN CONTRAST, A NEGATIVE (RED) NET AVERAGE FORWARD RETURN MEANS THAT THE REGION UNDERPERFORMED GLOBAL EQUITIES WHEN THE RELATIVE PRICE MOMENTUM WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A MEAN-REVERSION STRATEGY. SOURCE: MACROQUANT (REGION RUMBLE).

At the regional equity level, mean-reversion is quite common using daily data based on MSCI total return indices. We suspect that this result may be partly driven by a lack of liquidity for some of the securities underpinning these indices. In general, any noise in the underlying data will tend to bias the analysis towards showing mean reversion. Using monthly return data, return continuation tends to be the norm. This is especially true for emerging markets (Table 2).

Bonds: Buy 'Em When They Are Down

Bond yields show mean-reversion, especially over short-term horizons of several days or weeks. Since 2001, the TLT ETF has

delivered returns of 3 bps on days when it was down the prior day, while losing 2 bps on days when it was up the prior day (for a net return of 5 bps). The net return has averaged 7 bps since 2020.

A strategy of buying TLT and shorting SPY whenever the former underperformed the latter over the prior few days or weeks has been quite profitable. Since 2020, TLT has outperformed SPY by 14 basis points when TLT/SPY was down the prior day. In extreme cases where TLT/SPY fell by two standard deviations over the prior day, TLT outperformed SPY by an average of 0.66% over the following day. On the flipside, on days when TLT/SPY rose by two standard deviations, TLT underperformed SPY by an average of 0.63% on the following day.

Currencies and Commodities: The Trend is Your Friend

The US dollar is a momentum currency. Buying the greenback on strength and selling it on weakness has historically been a profitable strategy. Since 2001, the DXY index rose by 1 basis point over the subsequent day when its 200-day moving average z-score was above zero but fell by about 1 bp when the z-score was below zero. While this is not a huge difference, annualized over 12 months, it still amounts to a net return of 5.2%. Momentum was even stronger when assessed using monthly returns.

Of all the assets under consideration, commodities display the strongest price momentum. This is especially true for oil, gold, and silver.

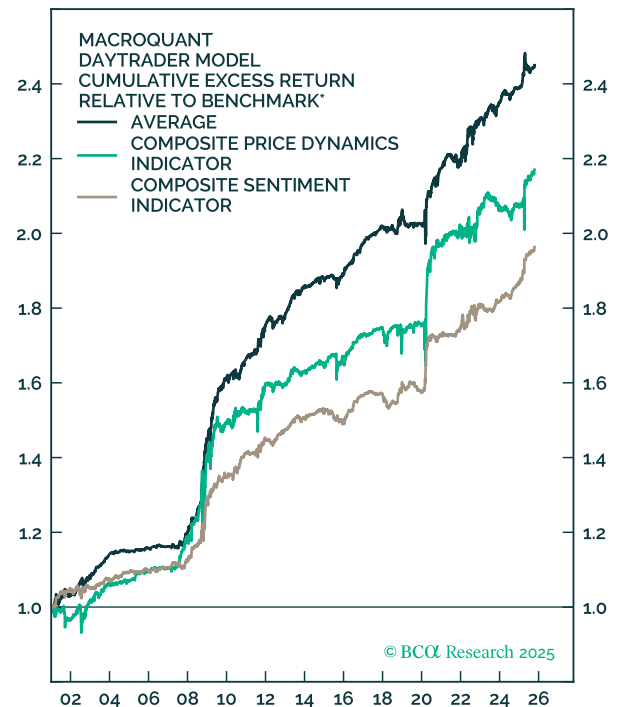
Unlike the other commodities, copper displayed mean reversion using daily returns. As with equity regions, we suspect that this may be partly due to liquidity/data noise issues, which can bias the conclusions towards mean reversion. Using less noisy monthly data, copper displays return continuation.

Not surprisingly, crypto prices display significant price momentum.

What Our Indicators Currently Say

Looking solely at price momentum, our indicators are currently saying that the S&P 500 was overbought as of Wednesday's close and hence vulnerable to a correction. Bond momentum is largely neutral. US dollar momentum is to the downside, which is negative for the greenback. The trend remains a friend for gold and silver but a foe for oil.

CHART 1
Combining Signals From Momentum
And Sentiment Data Amplifies Model
Performance



* SHOWN INDEXED TO JAN 4, 2001 = 1.0.
BENCHMARK DEFINED AS A STRATEGY THAT IS ON AVERAGE AS
BULLISH AS THE MACROQUANT DAYTRADER INDICATOR (50% EQUITY
HOLDING FOR AVERAGE, 51% EQUITY HOLDING FOR COMPOSITE
PRICE DYNAMICS INDICATOR, AND 44% EQUITY HOLDING FOR
COMPOSITE SENTIMENT INDICATOR).
SOURCE: MACROQUANT MULTIVERSE EDITION (DAYTRADER).

Of course, momentum is just one driver of asset prices. Our S&P 500 DayTrader model combines momentum readings with sentiment data to forecast returns. As **Chart 1** shows, combining both momentum and sentiment generates better results than looking at only one or the other.

For longer-term horizons of one month to five years, our [MacroQuant model](#) brings together hundreds of leading indicators for financial assets spanning the economy, monetary conditions, sentiment and positioning, earnings trends, and valuations.

TABLE 3
GIS PowerScores

	Z-Score			Net Return		
Asset Allocation (Absolute Return)	Daily	Weekly	Monthly	Daily	Weekly	Monthly
Equities (SPY)	0.11	0.15	0.18	0.29%	0.28%	0.13%
Bonds (TLT)	-0.48	0.03	0.42	0.24%	0.34%	0.25%
Cash (BIL)	0.16	0.16	0.16	0.02%	0.02%	0.02%
US Equity Sectors (Relative To S&P 500)	Daily	Weekly	Monthly	Daily	Weekly	Monthly
Communication Services (XLC)	0.62	0.66	0.69	0.16%	0.14%	0.13%
Consumer Discretionary (XLY)	0.26	0.21	0.15	0.09%	0.08%	0.07%
Consumer Staples (XLP)	-0.92	-1.05	-1.19	0.11%	0.10%	0.10%
Energy (XLE)	-1.96	-2.11	-2.27	0.18%	0.18%	0.16%
Financials (XLF)	0.09	-0.05	-0.20	0.19%	0.15%	0.13%
Health Care (XLV)	-0.77	-0.84	-0.91	0.06%	0.07%	0.06%
Industrials (XLI)	-0.16	-0.24	-0.31	0.06%	0.06%	0.06%
Information Technology (XLK)	0.35	0.49	0.62	0.06%	0.05%	0.05%
Materials (XLB)	-0.18	-0.27	-0.35	0.12%	0.12%	0.10%
Real Estate (XLRE)	-0.66	-0.71	-0.76	0.10%	0.09%	0.09%
Utilities (XLU)	-1.07	-1.11	-1.14	0.11%	0.09%	0.09%
Global Equity Regions (Relative To Global)	Daily	Weekly	Monthly	Daily	Weekly	Monthly
US (SPY)	0.24	0.22	0.19	0.08%	0.07%	0.06%
Euro Area (EZU)	-1.11	-1.10	-1.09	0.10%	0.08%	0.08%
UK (EWU)	-1.24	-1.35	-1.46	0.07%	0.05%	0.03%
Canada (EWC)	-0.74	-0.77	-0.79	0.06%	0.07%	0.06%
Australia (EWA)	-0.30	-0.29	-0.27	0.10%	0.08%	0.08%
Japan (EWJ)	0.25	0.38	0.51	0.13%	0.16%	0.19%
EM (EEM)	-0.17	-0.05	0.08	0.13%	0.12%	0.12%
Currencies And Commodities (Absolute Return)	Daily	Weekly	Monthly	Daily	Weekly	Monthly
DXU (UUP)	-0.67	-0.69	-0.70	0.12%	0.10%	0.11%
Oil (USO)	-1.11	-1.15	-1.19	0.50%	0.46%	0.47%
Copper (COPER)	0.32	0.33	0.34	0.49%	0.29%	0.22%
Gold (GLD)	1.62	1.47	1.32	0.21%	0.21%	0.21%

MACROQUANT POWERSCORES CAPTURE THE MODEL'S VIEW ON DIFFERENT ASSETS OVER VARIOUS TIME HORIZONS. POSITIVE SCORES IMPLY A HIGHER-THAN-AVERAGE PROSPECTIVE RETURN, WHEREAS NEGATIVE SCORES IMPLY A LOWER-THAN-AVERAGE RETURN. THE DAILY SCORES ARE FAST-MOVING AND THUS OPTIMIZED FOR HOLDING HORIZONS OF UP TO 15 DAYS; WEEKLY SCORES ARE SLOWER MOVING AND THUS APPROPRIATE FOR HOLDING HORIZONS OF UP TO 13 WEEKS; MONTHLY SCORES TYPICALLY EXHIBIT LITTLE CHANGE ON A DAY-TO-DAY BASIS AND HENCE ARE BETTER SUITED FOR HORIZONS OF UP TO 12 MONTHS. THE "NET RETURN" IS THE DIFFERENCE IN THE AVERAGE SUBSEQUENT ONE-DAY RETURN BETWEEN A BULLISH SIGNAL (POWERSCORE ABOVE 0.5) AND A BEARISH SIGNAL (POWERSCORE BELOW -0.5).

Our newly revised GIS PowerScores combine the output from our short-term models and our longer-term models. We assign different weights to the models depending on the time horizon over which projected returns are assessed.

Table 3 shows where the PowerScores stand today. Positive scores imply a higher-than-average prospective return, whereas negative scores imply a lower-than-average return. The daily scores are fast-moving and thus optimized for holding horizons of up to 15 days; weekly scores are slower-moving and thus appropriate for holding horizons of up to 13 weeks; monthly scores typically exhibit little change on a day-to-day basis and hence are better suited for horizons of up to 12 months.

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Appendix

APPENDIX TABLE 1

Net Forward Average Return To A Trend-Following Strategy Based On The Z-Score Of Past One-Day Return (2001-To-2025)

Z-Score Threshold:	Below 0, Above 0	Below -0.5, Above 0.5	Below -1, Above 1	Below -1.5, Above 1.5	Below -2, Above 2
S&P 500 (SPY)	-0.12	-0.20	-0.35	-0.50	-0.83
Russell 2000 (IWM)	-0.10	-0.17	-0.26	-0.51	-0.90
S&P 600 (IJR)	-0.08	-0.14	-0.24	-0.44	-0.89
Value (IVE)	-0.10	-0.14	-0.23	-0.38	-0.69
Growth (IVW)	-0.14	-0.21	-0.41	-0.63	-1.14
Momentum (MTUM)	-0.09	-0.12	-0.22	-0.46	-0.85
Quality (QUAL)	-0.11	-0.21	-0.34	-0.46	-0.84
Russell 2000/S&P 500 (IWM/SPY)	-0.01	-0.06	-0.07	-0.04	-0.04
S&P 600/S&P 500 (IJR/SPY)	0.00	-0.05	-0.07	-0.12	-0.19
Value/S&P 500 (IVE/SPY)	0.01	0.02	0.03	0.02	0.02
Growth/S&P 500 (IVW/SPY)	0.01	0.01	0.03	0.01	-0.01
Momentum/S&P 500 (MTUM/SPY)	0.05	0.09	0.10	0.12	0.19
Quality/S&P 500 (QUAL/SPY)	0.00	0.01	-0.02	-0.03	-0.03
Communication Services/S&P 500 (XLC/SPY)	-0.05	-0.05	-0.07	-0.06	-0.07
Consumer Discretionary/S&P 500 (XLY/SPY)	0.02	0.04	0.01	-0.04	-0.05
Consumer Staples/S&P 500 (XLP/SPY)	0.03	0.00	-0.07	-0.04	-0.09
Energy/S&P 500 (XLE/SPY)	0.02	0.07	0.06	0.04	0.03
Financials/S&P 500 (XLF/SPY)	-0.07	-0.14	-0.28	-0.49	-0.64
Health Care/S&P 500 (XLV/SPY)	0.04	0.06	0.08	0.01	-0.22
Industrials/S&P 500 (XLI/SPY)	0.02	0.01	0.01	0.07	-0.02
IT/S&P 500 (XLK/SPY)	-0.02	0.00	0.05	-0.05	0.24
Materials/S&P 500 (XLB/SPY)	0.01	-0.03	-0.01	0.13	0.18
Real Estate/S&P 500 (XLRE/SPY)	-0.15	-0.32	-0.58	-1.49	-2.19
Utilities/S&P 500 (XLU/SPY)	-0.04	-0.03	-0.07	-0.11	-0.18
US/Global (SPY/ACWI)	-0.21	-0.43	-0.75	-1.15	-1.35
Euro Area/Global (EZU/ACWI)	-0.27	-0.48	-0.73	-1.01	-1.44
UK/Global (EWU/ACWI)	-0.24	-0.43	-0.85	-1.32	-1.71
Canada/Global (EWC/ACWI)	-0.20	-0.38	-0.68	-1.25	-1.36
Australia/Global (EWA/ACWI)	-0.26	-0.53	-1.01	-1.60	-2.56
Japan/Global (EWJ/ACWI)	-0.11	-0.13	-0.25	-0.42	-0.88
EM/Global (EEM/ACWI)	-0.23	-0.44	-0.84	-1.46	-2.18
China/Global (MCHI/ACWI)	-0.06	-0.13	-0.20	-0.18	-0.17
Brazil/Global (EWZ/ACWI)	-0.13	-0.15	-0.43	-1.03	-1.52
India/Global (INDA/ACWI)	-0.06	-0.06	-0.20	-0.59	-1.19
Long-term Bond (TLT)	-0.05	-0.06	-0.12	-0.07	-0.16
Long-term Bond/S&P 500 (TLT/SPY)	-0.14	-0.24	-0.59	-0.81	-1.29
High-Yield Corporate (HYG)	0.04	0.07	0.08	0.27	0.14
High-Yield Corporate/S&P 500 (HYG/SPY)	-0.12	-0.30	-0.50	-0.77	-1.35
DXY	-0.01	-0.01	0.00	-0.01	0.01
USD/EUR	-0.02	-0.02	0.05	0.04	0.15
USD/GBP	0.01	0.01	0.01	0.07	0.15
USD/CHF	-0.03	-0.01	-0.01	0.04	0.01
USD/SEK	-0.01	0.00	0.00	-0.02	-0.02
USD/NOK	0.00	0.03	0.03	0.02	0.11
USD/CAD	-0.02	-0.01	-0.01	0.06	0.18
USD/AUD	0.00	-0.02	-0.03	0.03	0.11
USD/NZD	0.00	0.01	0.07	0.11	0.14
USD/JPY	0.00	0.00	-0.04	-0.10	-0.13
USD/CNY	-0.01	-0.02	-0.03	-0.03	-0.05
USD/BRL	0.05	0.05	0.11	0.04	0.03
USD/INR	0.00	-0.01	-0.02	-0.04	-0.06
USD/MXN	0.03	0.03	0.04	0.13	0.22
EUR/CHF	0.00	0.01	0.04	0.04	0.11
EUR/GBP	-0.02	-0.01	-0.02	-0.04	0.06
EUR/SEK	-0.01	-0.02	-0.03	-0.02	-0.07
EUR/JPY	-0.01	0.00	0.01	-0.02	0.02
BTC	0.30	0.11	0.45	-0.64	-1.19
ETH	-0.19	-0.22	-0.33	-1.73	-2.59
ETH/BTC	0.03	0.21	-0.40	-0.65	0.63
USO	-0.01	0.00	0.13	0.26	-0.26
CPER	-0.12	-0.17	-0.22	-0.30	-0.46
SLV	0.01	-0.03	0.01	0.42	1.16
GLD	-0.04	-0.03	-0.04	0.07	0.13
USO/CPER	-0.05	-0.04	0.08	0.23	-0.11
USO/GLD	0.01	-0.01	0.05	0.04	-0.17
CPER/GLD	-0.12	-0.21	-0.32	-0.27	-0.48
SLV/GLD	0.04	-0.02	0.03	0.08	-0.05

NOTE: A POSITIVE (GREEN) NET AVERAGE FORWARD RETURN MEANS THAT THE SUBSEQUENT DAY'S RETURN WAS ABOVE-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A TREND-FOLLOWING STRATEGY. IN CONTRAST, A NEGATIVE (RED) NET AVERAGE FORWARD RETURN MEANS THAT THE RETURN WAS BELOW-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A MEAN-REVERSION STRATEGY.

APPENDIX TABLE 2

Net Forward Average Return To A Trend-Following Strategy Based On The Z-Score Of Past One-Day Return (2020-To-2025)

Z-Score Threshold:	Below 0, Above 0	Below -0.5, Above 0.5	Below -1, Above 1	Below -1.5, Above 1.5	Below -2, Above 2
S&P 500 (SPY)	-0.13	-0.25	-0.51	-0.91	-1.65
Russell 2000 (IWM)	-0.14	-0.18	-0.31	-0.67	-1.15
S&P 600 (IJR)	-0.16	-0.15	-0.40	-0.57	-1.17
Value (IVE)	-0.05	-0.10	-0.23	-0.54	-1.97
Growth (IVW)	-0.26	-0.25	-0.53	-0.88	-1.08
Momentum (MTUM)	-0.19	-0.18	-0.41	-0.59	-1.17
Quality (QUAL)	-0.11	-0.24	-0.41	-0.70	-1.11
Russell 2000/S&P 500 (IWM/SPY)	0.02	-0.03	-0.03	0.08	0.19
S&P 600/S&P 500 (IJR/SPY)	0.03	-0.07	-0.01	-0.02	-0.06
Value/S&P 500 (IVE/SPY)	0.02	0.01	0.03	0.04	0.06
Growth/S&P 500 (IVW/SPY)	0.02	0.00	0.02	0.02	-0.02
Momentum/S&P 500 (MTUM/SPY)	0.04	0.08	0.06	0.07	0.16
Quality/S&P 500 (QUAL/SPY)	-0.01	-0.01	-0.03	-0.02	0.01
Communication Services/S&P 500 (XLC/SPY)	-0.05	-0.05	-0.09	-0.06	-0.06
Consumer Discretionary/S&P 500 (XLY/SPY)	0.02	0.02	-0.03	-0.16	-0.30
Consumer Staples/S&P 500 (XLP/SPY)	0.01	-0.02	-0.04	0.06	0.05
Energy/S&P 500 (XLE/SPY)	0.11	0.21	0.23	0.27	0.18
Financials/S&P 500 (XLF/SPY)	-0.04	-0.03	-0.01	-0.07	-0.09
Health Care/S&P 500 (XLV/SPY)	0.00	-0.02	0.00	-0.02	-0.12
Industrials/S&P 500 (XLI/SPY)	0.02	0.01	-0.02	0.07	0.01
IT/S&P 500 (XLK/SPY)	-0.08	-0.07	-0.02	-0.21	-0.56
Materials/S&P 500 (XLB/SPY)	0.02	0.01	0.04	0.15	0.19
Real Estate/S&P 500 (XLRE/SPY)	-0.13	-0.15	-0.01	-0.11	0.01
Utilities/S&P 500 (XLU/SPY)	-0.10	-0.08	-0.12	-0.02	-0.10
US/Global (SPY/ACWI)	-0.20	-0.37	-0.71	-1.09	-1.85
Euro Area/Global (EZU/ACWI)	-0.14	-0.24	-0.20	0.18	0.45
UK/Global (EWU/ACWI)	-0.12	-0.14	-0.22	-0.31	-0.37
Canada/Global (EWC/ACWI)	-0.04	-0.10	-0.16	-0.35	-0.61
Australia/Global (EWA/ACWI)	-0.28	-0.60	-1.17	-2.26	-3.19
Japan/Global (EWJ/ACWI)	-0.05	0.02	-0.05	-0.22	-0.01
EM/Global (EEM/ACWI)	-0.15	-0.27	-0.37	-0.59	-1.54
China/Global (MCHI/ACWI)	-0.11	-0.17	-0.29	-0.28	-0.42
Brazil/Global (EWZ/ACWI)	-0.23	-0.22	-0.48	-1.18	-2.04
India/Global (INDA/ACWI)	-0.05	-0.07	-0.37	-1.30	-2.56
Long-term Bond (TLT)	-0.07	-0.03	-0.09	-0.10	-0.11
Long-term Bond/S&P 500 (TLT/SPY)	-0.17	-0.25	-0.69	-1.56	-2.86
High-Yield Corporate (HYG)	-0.02	0.00	0.09	0.25	-0.01
High-Yield Corporate/S&P 500 (HYG/SPY)	-0.09	-0.11	-0.43	-0.70	-1.29
DXY	0.02	0.04	0.05	0.04	0.10
USD/EUR	0.02	0.00	0.03	0.06	-0.06
USD/GBP	-0.01	0.02	-0.01	-0.04	0.07
USD/CHF	0.02	0.08	0.05	0.04	0.26
USD/SEK	-0.02	-0.02	-0.08	-0.06	-0.10
USD/NOK	-0.03	-0.02	0.08	0.22	0.33
USD/CAD	-0.03	-0.01	0.02	0.07	0.14
USD/AUD	-0.06	-0.10	-0.03	0.13	0.24
USD/NZD	-0.06	-0.10	-0.06	0.09	0.18
USD/JPY	0.00	0.03	-0.02	-0.10	-0.20
USD/CNY	-0.01	-0.03	-0.04	-0.05	-0.08
USD/BRL	0.03	0.02	-0.01	-0.21	-0.48
USD/INR	0.00	-0.03	-0.13	-0.16	-0.22
USD/MXN	0.00	-0.02	0.04	0.15	0.18
EUR/CHF	-0.02	-0.04	-0.07	-0.19	-0.26
EUR/GBP	-0.03	-0.04	-0.08	-0.14	-0.04
EUR/SEK	-0.05	-0.09	-0.19	-0.24	-0.35
EUR/JPY	0.00	-0.06	-0.06	-0.17	-0.36
BTC	-0.09	-0.27	-0.35	-1.91	-1.75
ETH	-0.22	-0.20	-0.57	-2.17	-3.83
ETH/BTC	-0.04	0.13	-0.46	-1.10	0.23
USO	0.08	0.33	1.07	1.55	1.52
CPER	-0.12	-0.16	-0.18	-0.17	-0.39
SLV	-0.06	0.03	0.13	0.24	1.22
GLD	0.01	0.01	0.13	0.28	0.45
USO/CPER	-0.02	0.12	0.58	1.30	1.28
USO/GLD	0.15	0.28	0.79	1.25	1.05
CPER/GLD	-0.15	-0.25	-0.33	-0.22	-0.68
SLV/GLD	0.07	0.05	0.13	0.24	0.14

NOTE: A POSITIVE (GREEN) NET AVERAGE FORWARD RETURN MEANS THAT THE SUBSEQUENT DAY'S RETURN WAS ABOVE-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A TREND-FOLLOWING STRATEGY. IN CONTRAST, A NEGATIVE (RED) NET AVERAGE FORWARD RETURN MEANS THAT THE RETURN WAS BELOW-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A MEAN-REVERSION STRATEGY.

APPENDIX TABLE 3

Net Forward Average Return To A Trend-Following Strategy Based On The Z-Score Of Price Deviation From 10-Day Moving Average (2001-To-2025)

Z-Score Threshold:	Below 0, Above 0	Below -0.5, Above 0.5	Below -1, Above 1	Below -1.5, Above 1.5	Below -2, Above 2
S&P 500 (SPY)	-0.07	-0.13	-0.22	-0.41	-0.61
Russell 2000 (IWM)	-0.08	-0.12	-0.25	-0.42	-0.70
S&P 600 (IJR)	-0.08	-0.12	-0.17	-0.42	-0.74
Value (IVE)	-0.05	-0.13	-0.15	-0.33	-0.57
Growth (IVW)	-0.05	-0.14	-0.22	-0.43	-1.06
Momentum (MTUM)	-0.10	-0.12	-0.10	-0.18	-0.91
Quality (QUAL)	-0.06	-0.15	-0.23	-0.33	-0.57
Russell 2000/S&P 500 (IWM/SPY)	-0.06	-0.08	-0.11	-0.17	-0.14
S&P 600/S&P 500 (IJR/SPY)	-0.04	-0.08	-0.14	-0.13	-0.22
Value/S&P 500 (IVE/SPY)	0.01	0.01	-0.01	-0.02	-0.09
Growth/S&P 500 (IVW/SPY)	0.01	0.01	-0.01	-0.03	-0.11
Momentum/S&P 500 (MTUM/SPY)	0.02	0.03	0.00	-0.10	-0.23
Quality/S&P 500 (QUAL/SPY)	0.00	0.00	-0.01	-0.05	-0.07
Communication Services/S&P 500 (XLC/SPY)	-0.06	-0.10	-0.16	-0.15	-0.09
Consumer Discretionary/S&P 500 (XLY/SPY)	-0.01	-0.02	-0.04	-0.02	-0.07
Consumer Staples/S&P 500 (XLP/SPY)	0.01	-0.01	-0.13	-0.40	-0.74
Energy/S&P 500 (XLE/SPY)	0.04	0.06	0.00	-0.11	-0.18
Financials/S&P 500 (XLF/SPY)	-0.06	-0.15	-0.31	-0.47	-0.65
Health Care/S&P 500 (XLV/SPY)	0.03	0.01	0.00	-0.08	-0.29
Industrials/S&P 500 (XLI/SPY)	0.02	0.02	-0.04	-0.10	-0.24
IT/S&P 500 (XLK/SPY)	0.00	-0.07	-0.21	-0.39	-1.06
Materials/S&P 500 (XLB/SPY)	-0.04	-0.03	-0.08	-0.23	-0.10
Real Estate/S&P 500 (XLRE/SPY)	-0.10	-0.20	-0.43	-1.06	-2.01
Utilities/S&P 500 (XLU/SPY)	-0.03	-0.04	-0.13	-0.30	-0.43
US/Global (SPY/ACWI)	-0.21	-0.37	-0.59	-0.94	-1.38
Euro Area/Global (EZU/ACWI)	-0.22	-0.36	-0.57	-0.71	-0.87
UK/Global (EWU/ACWI)	-0.19	-0.34	-0.53	-0.84	-1.28
Canada/Global (EWC/ACWI)	-0.17	-0.27	-0.32	-0.61	-1.16
Australia/Global (EWA/ACWI)	-0.24	-0.40	-0.69	-1.14	-1.92
Japan/Global (EWJ/ACWI)	-0.07	-0.13	-0.24	-0.34	-0.81
EM/Global (EEM/ACWI)	-0.23	-0.41	-0.74	-1.53	-2.53
China/Global (MCHI/ACWI)	-0.02	-0.01	0.00	-0.09	-0.18
Brazil/Global (EWZ/ACWI)	-0.13	-0.21	-0.46	-0.76	-1.19
India/Global (INDA/ACWI)	-0.08	-0.09	-0.18	-0.27	-0.75
Long-term Bond (TLT)	-0.04	-0.08	-0.14	-0.25	-0.49
Long-term Bond/S&P 500 (TLT/SPY)	-0.14	-0.23	-0.44	-0.59	-0.93
High-Yield Corporate (HYG)	0.03	0.04	0.10	0.05	-0.17
High-Yield Corporate/S&P 500 (HYG/SPY)	-0.10	-0.20	-0.39	-0.81	-1.60
DXE	0.00	0.00	-0.02	-0.02	-0.09
USD/EUR	-0.01	0.01	-0.01	-0.01	-0.02
USD/GBP	-0.02	-0.02	-0.03	0.05	0.00
USD/CHF	-0.01	-0.01	0.01	0.08	0.12
USD/SEK	-0.01	-0.02	-0.06	-0.13	-0.24
USD/NOK	-0.01	-0.01	-0.04	-0.13	-0.20
USD/CAD	-0.02	-0.02	-0.03	0.03	0.06
USD/AUD	0.00	0.01	-0.06	-0.07	-0.27
USD/NZD	0.00	-0.01	0.01	-0.06	-0.06
USD/JPY	-0.02	0.00	-0.04	-0.04	-0.15
USD/CNY	-0.01	-0.01	-0.01	0.00	0.01
USD/BRL	0.03	0.00	-0.05	-0.09	-0.26
USD/INR	0.01	0.02	0.03	0.02	-0.01
USD/MXN	0.00	0.01	-0.03	-0.10	-0.05
EUR/CHF	-0.01	0.00	0.00	0.02	-0.06
EUR/GBP	-0.01	-0.03	-0.04	-0.10	-0.06
EUR/SEK	-0.03	-0.03	-0.06	-0.10	-0.18
EUR/JPY	0.00	-0.01	-0.04	-0.06	-0.21
BTC	0.44	0.74	0.59	-0.65	-0.72
ETH	0.43	0.47	-0.52	-1.89	-5.41
ETH/BTC	0.31	0.53	-0.49	-1.20	...
USO	0.07	0.11	0.15	-0.05	-0.20
CPER	-0.10	-0.13	-0.14	-0.12	-0.27
SLV	0.05	0.11	0.09	-0.09	-0.29
GLD	0.03	0.01	-0.06	-0.10	-0.20
USO/CPER	0.04	0.00	0.02	-0.26	-0.12
USO/GLD	0.17	0.21	0.24	0.17	-0.22
CPER/GLD	-0.14	-0.16	-0.22	-0.38	-0.62
SLV/GLD	0.00	0.07	0.13	0.08	0.26

NOTE: A POSITIVE (GREEN) NET AVERAGE FORWARD RETURN MEANS THAT THE SUBSEQUENT DAY'S RETURN WAS ABOVE-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A TREND-FOLLOWING STRATEGY. IN CONTRAST, A NEGATIVE (RED) NET AVERAGE FORWARD RETURN MEANS THAT THE RETURN WAS BELOW-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A MEAN-REVERSION STRATEGY.

APPENDIX TABLE 4

Net Forward Average Return To A Trend-Following Strategy Based On The Z-Score Of Price Deviation From 10-Day Moving Average (2020-To-2025)

Z-Score Threshold:	Below 0, Above 0	Below -0.5, Above 0.5	Below -1, Above 1	Below -1.5, Above 1.5	Below -2, Above 2
S&P 500 (SPY)	-0.04	-0.05	-0.18	-0.20	-0.37
Russell 2000 (IWM)	-0.01	-0.04	-0.11	-0.22	-0.04
S&P 600 (IJR)	-0.10	-0.07	-0.08	-0.16	-0.29
Value (IVE)	-0.01	-0.03	0.03	0.13	0.04
Growth (IVW)	0.01	-0.07	-0.12	-0.25	-0.64
Momentum (MTUM)	-0.06	0.05	-0.07	-0.06	-0.49
Quality (QUAL)	-0.02	-0.09	-0.07	-0.28	-0.41
Russell 2000/S&P 500 (IWM/SPY)	-0.09	-0.08	0.01	-0.02	0.04
S&P 600/S&P 500 (IJR/SPY)	-0.06	-0.09	-0.09	0.05	-0.06
Value/S&P 500 (IVE/SPY)	0.01	0.03	0.00	0.01	-0.07
Growth/S&P 500 (IVW/SPY)	0.01	0.02	0.00	0.02	-0.08
Momentum/S&P 500 (MTUM/SPY)	0.02	0.01	0.01	-0.02	-0.23
Quality/S&P 500 (QUAL/SPY)	0.00	-0.02	-0.03	-0.07	-0.06
Communication Services/S&P 500 (XLC/SPY)	-0.06	-0.10	-0.15	-0.15	-0.09
Consumer Discretionary/S&P 500 (XLY/SPY)	0.01	-0.01	0.00	-0.01	-0.10
Consumer Staples/S&P 500 (XLP/SPY)	-0.03	-0.02	-0.13	-0.27	-0.77
Energy/S&P 500 (XLE/SPY)	0.18	0.24	0.27	0.29	0.21
Financials/S&P 500 (XLF/SPY)	0.00	-0.03	-0.07	-0.18	-0.31
Health Care/S&P 500 (XLV/SPY)	0.04	0.03	0.01	0.02	-0.26
Industrials/S&P 500 (XLI/SPY)	0.01	-0.02	0.00	-0.10	-0.25
IT/S&P 500 (XLK/SPY)	0.01	-0.05	-0.22	-0.50	-0.96
Materials/S&P 500 (XLB/SPY)	-0.02	-0.06	0.04	-0.01	-0.01
Real Estate/S&P 500 (XLRE/SPY)	-0.07	-0.12	-0.15	-0.41	-0.96
Utilities/S&P 500 (XLU/SPY)	-0.07	-0.09	-0.21	-0.37	-0.65
US/Global (SPY/ACWI)	-0.18	-0.35	-0.54	-0.76	-1.38
Euro Area/Global (EZU/ACWI)	-0.07	-0.07	-0.12	0.04	0.23
UK/Global (EWU/ACWI)	-0.04	-0.04	-0.09	-0.14	-0.22
Canada/Global (EWC/ACWI)	-0.05	-0.09	-0.12	-0.30	-0.85
Australia/Global (EWA/ACWI)	-0.20	-0.32	-0.43	-1.07	-1.74
Japan/Global (EWJ/ACWI)	-0.01	-0.02	-0.11	-0.15	-0.23
EM/Global (EEM/ACWI)	-0.06	-0.20	-0.28	-0.41	-0.97
China/Global (MCHI/ACWI)	0.07	0.08	0.14	-0.10	-0.31
Brazil/Global (EWZ/ACWI)	-0.02	-0.05	-0.05	-0.28	-0.68
India/Global (INDA/ACWI)	-0.06	-0.11	-0.34	-1.06	-1.99
Long-term Bond (TLT)	0.01	-0.07	-0.21	-0.54	-0.80
Long-term Bond/S&P 500 (TLT/SPY)	-0.23	-0.33	-0.68	-0.87	-0.78
High-Yield Corporate (HYG)	-0.02	-0.02	0.05	-0.10	-0.07
High-Yield Corporate/S&P 500 (HYG/SPY)	-0.06	-0.13	-0.18	-0.32	-1.16
DXY	0.01	-0.01	-0.06	-0.03	-0.07
USD/EUR	-0.02	-0.02	-0.09	0.02	0.00
USD/GBP	-0.03	-0.04	-0.13	0.03	-0.14
USD/CHF	-0.02	0.00	0.06	0.05	0.13
USD/SEK	-0.02	-0.08	-0.05	-0.15	-0.14
USD/NOK	-0.06	-0.03	-0.07	0.01	0.17
USD/CAD	-0.03	-0.03	-0.12	0.14	0.17
USD/AUD	-0.04	-0.08	-0.14	-0.06	0.11
USD/NZD	-0.05	-0.05	-0.06	-0.03	0.45
USD/JPY	-0.02	-0.02	-0.09	-0.08	-0.13
USD/CNY	-0.01	-0.02	-0.04	-0.04	-0.03
USD/BRL	0.03	0.01	-0.05	-0.40	-0.42
USD/INR	-0.01	-0.02	-0.05	-0.08	-0.44
USD/MXN	-0.01	0.00	-0.06	-0.19	0.01
EUR/CHF	0.00	0.00	-0.04	-0.14	-0.40
EUR/GBP	-0.04	-0.08	-0.10	-0.09	0.33
EUR/SEK	-0.03	-0.04	-0.03	-0.15	-0.25
EUR/JPY	-0.03	-0.09	-0.18	-0.22	-0.05
BTC	0.14	0.42	0.19	-1.23	-1.25
ETH	0.24	0.33	-0.90	-2.76	-5.20
ETH/BTC	0.23	0.42	-0.28	-1.25	...
USO	0.16	0.19	0.23	0.33	1.87
CPER	-0.05	-0.08	-0.01	0.07	0.14
SLV	0.05	0.13	0.13	-0.18	0.39
GLD	0.04	-0.01	-0.19	-0.13	-0.12
USO/CPER	0.06	-0.02	0.18	0.18	0.12
USO/GLD	0.28	0.23	0.41	0.57	0.18
CPER/GLD	-0.21	-0.27	-0.18	-0.20	0.06
SLV/GLD	0.06	0.13	0.14	0.13	0.44

NOTE: A POSITIVE (GREEN) NET AVERAGE FORWARD RETURN MEANS THAT THE SUBSEQUENT DAY'S RETURN WAS ABOVE-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A TREND-FOLLOWING STRATEGY. IN CONTRAST, A NEGATIVE (RED) NET AVERAGE FORWARD RETURN MEANS THAT THE RETURN WAS BELOW-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A MEAN-REVERSION STRATEGY.

APPENDIX TABLE 5

**Net Forward Average Return To A Trend-Following Strategy Based On The Z-Score Of Price
Deviation From 50-Day Moving Average (2001-To-2025)**

Z-Score Threshold:	Below 0, Above 0	Below -0.5, Above 0.5	Below -1, Above 1	Below -1.5, Above 1.5	Below -2, Above 2
S&P 500 (SPY)	-0.04	-0.06	-0.17	-0.31	-0.57
Russell 2000 (IWM)	-0.03	-0.05	-0.12	-0.19	-0.27
S&P 600 (IJR)	-0.05	-0.08	-0.10	-0.12	-0.08
Value (IVE)	-0.03	-0.05	-0.10	-0.28	-0.41
Growth (IVW)	-0.02	-0.08	-0.17	-0.41	-0.38
Momentum (MTUM)	-0.02	-0.10	-0.19	-0.31	-0.79
Quality (QUAL)	-0.04	-0.07	-0.16	-0.35	-0.60
Russell 2000/S&P 500 (IWM/SPY)	-0.01	-0.05	-0.10	-0.10	-0.21
S&P 600/S&P 500 (IJR/SPY)	-0.02	-0.05	-0.08	-0.11	-0.12
Value/S&P 500 (IVE/SPY)	0.01	0.01	-0.02	0.00	-0.03
Growth/S&P 500 (IVW/SPY)	0.01	0.01	-0.01	0.02	-0.09
Momentum/S&P 500 (MTUM/SPY)	0.02	-0.01	0.01	-0.07	-0.10
Quality/S&P 500 (QUAL/SPY)	0.00	0.00	0.00	0.00	-0.02
Communication Services/S&P 500 (XLC/SPY)	-0.06	-0.06	-0.01	-0.02	-0.05
Consumer Discretionary/S&P 500 (XLY/SPY)	-0.01	0.00	0.02	-0.03	-0.17
Consumer Staples/S&P 500 (XLP/SPY)	0.02	-0.01	-0.11	-0.20	-0.49
Energy/S&P 500 (XLE/SPY)	0.04	0.04	0.00	-0.01	-0.02
Financials/S&P 500 (XLF/SPY)	-0.02	-0.05	-0.18	-0.29	-0.27
Health Care/S&P 500 (XLV/SPY)	0.01	0.01	-0.01	-0.19	-0.22
Industrials/S&P 500 (XLI/SPY)	0.00	-0.02	-0.06	-0.03	-0.23
IT/S&P 500 (XLK/SPY)	-0.04	-0.13	-0.46	-1.08	...
Materials/S&P 500 (XLB/SPY)	-0.03	-0.09	-0.24	-0.29	-0.58
Real Estate/S&P 500 (XLRE/SPY)	-0.07	-0.10	-0.30	-0.60	-1.30
Utilities/S&P 500 (XLU/SPY)	0.01	0.00	-0.10	-0.43	-0.82
US/Global (SPY/ACWI)	-0.10	-0.16	-0.26	-0.49	-0.60
Euro Area/Global (EZU/ACWI)	-0.09	-0.14	-0.19	-0.29	-0.59
UK/Global (EWU/ACWI)	-0.15	-0.20	-0.32	-0.41	-0.50
Canada/Global (EWC/ACWI)	-0.09	-0.11	-0.20	-0.46	-0.79
Australia/Global (EWA/ACWI)	-0.12	-0.22	-0.29	-0.44	-0.86
Japan/Global (EWJ/ACWI)	-0.05	-0.08	-0.15	-0.35	-0.70
EM/Global (EEM/ACWI)	-0.14	-0.23	-0.37	-0.67	-1.05
China/Global (MCHI/ACWI)	0.04	0.03	0.00	-0.01	-0.26
Brazil/Global (EWZ/ACWI)	0.05	0.04	-0.07	-0.20	-0.75
India/Global (INDA/ACWI)	-0.05	-0.07	-0.08	-0.09	-0.36
Long-term Bond (TLT)	-0.02	-0.01	-0.03	-0.06	-0.11
Long-term Bond/S&P 500 (TLT/SPY)	-0.02	-0.05	-0.19	-0.26	-0.38
High-Yield Corporate (HYG)	0.01	0.01	0.05	0.04	-0.07
High-Yield Corporate/S&P 500 (HYG/SPY)	-0.06	-0.14	-0.23	-0.66	-2.04
DXY	0.01	0.04	0.01	-0.01	-0.07
USD/EUR	0.01	0.03	0.02	-0.07	-0.06
USD/GBP	0.00	-0.03	-0.01	0.06	0.05
USD/CHF	-0.01	-0.03	-0.06	-0.09	-0.02
USD/SEK	-0.01	0.00	0.00	-0.04	-0.09
USD/NOK	-0.01	-0.02	-0.04	-0.01	0.01
USD/CAD	-0.02	-0.02	-0.01	0.04	0.04
USD/AUD	0.00	0.04	0.00	-0.01	-0.05
USD/NZD	0.01	0.00	-0.03	-0.08	-0.23
USD/JPY	-0.01	0.01	0.02	0.02	-0.07
USD/CNY	0.01	0.02	0.03	0.04	0.06
USD/BRL	0.05	0.05	0.08	0.09	0.20
USD/INR	0.01	0.04	0.06	0.04	0.06
USD/MXN	0.01	0.01	-0.02	0.00	0.01
EUR/CHF	-0.02	-0.02	-0.04	-0.07	-0.08
EUR/GBP	-0.02	-0.03	-0.04	-0.06	-0.18
EUR/SEK	-0.02	-0.04	-0.05	-0.11	-0.08
EUR/JPY	-0.02	-0.03	-0.03	0.02	0.04
BTC	0.50	0.64	0.55	-0.06	-2.12
ETH	0.33	0.42	0.52	-1.92	...
ETH/BTC	0.14	0.07	0.54	...	...
USO	0.07	0.20	0.40	0.38	0.37
CPER	-0.07	-0.16	-0.12	-0.09	-0.31
SLV	0.10	0.08	-0.08	-0.43	-0.34
GLD	0.02	-0.01	-0.09	-0.16	-0.76
USO/CPER	-0.02	0.06	0.15	-0.06	-0.20
USO/GLD	0.16	0.27	0.54	0.55	0.59
CPER/GLD	-0.10	-0.17	-0.18	-0.25	-0.21
SLV/GLD	0.05	0.02	0.09	0.16	0.01

NOTE: A POSITIVE (GREEN) NET AVERAGE FORWARD RETURN MEANS THAT THE SUBSEQUENT DAY'S RETURN WAS ABOVE-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A TREND-FOLLOWING STRATEGY. IN CONTRAST, A NEGATIVE (RED) NET AVERAGE FORWARD RETURN MEANS THAT THE RETURN WAS BELOW-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A MEAN-REVERSION STRATEGY.

APPENDIX TABLE 6

Net Forward Average Return To A Trend-Following Strategy Based On The Z-Score Of Price Deviation From 50-Day Moving Average (2020-To-2025)

Z-Score Threshold:	Below 0, Above 0	Below -0.5, Above 0.5	Below -1, Above 1	Below -1.5, Above 1.5	Below -2, Above 2
S&P 500 (SPY)	-0.02	-0.04	-0.12	-0.31	-0.63
Russell 2000 (IWM)	0.03	0.00	-0.09	-0.02	-0.47
S&P 600 (IJR)	-0.01	-0.07	-0.06	-0.03	0.17
Value (IVE)	-0.03	-0.05	-0.11	-0.13	-0.28
Growth (IVW)	0.03	0.00	-0.12	-0.23	-0.34
Momentum (MTUM)	-0.02	-0.04	-0.20	-0.30	-0.85
Quality (QUAL)	-0.08	-0.02	-0.11	-0.19	-0.65
Russell 2000/S&P 500 (IWM/SPY)	0.01	-0.06	0.00	0.03	0.05
S&P 600/S&P 500 (IJR/SPY)	-0.01	-0.03	-0.01	0.00	0.04
Value/S&P 500 (IVE/SPY)	0.04	0.04	0.00	0.04	-0.02
Growth/S&P 500 (IVW/SPY)	0.04	0.04	0.02	0.04	-0.09
Momentum/S&P 500 (MTUM/SPY)	0.00	-0.05	0.02	0.00	-0.02
Quality/S&P 500 (QUAL/SPY)	0.00	-0.02	-0.02	-0.02	...
Communication Services/S&P 500 (XLC/SPY)	-0.05	-0.05	-0.01	-0.02	-0.05
Consumer Discretionary/S&P 500 (XLY/SPY)	-0.01	0.01	0.08	0.01	-0.09
Consumer Staples/S&P 500 (XLP/SPY)	0.00	-0.01	-0.01	-0.16	-0.37
Energy/S&P 500 (XLE/SPY)	0.20	0.23	0.23	0.24	0.29
Financials/S&P 500 (XLF/SPY)	-0.01	0.02	-0.05	-0.16	-0.37
Health Care/S&P 500 (XLV/SPY)	-0.01	0.01	0.00	-0.15	-0.20
Industrials/S&P 500 (XLI/SPY)	0.01	-0.03	-0.08	-0.03	-0.22
IT/S&P 500 (XLK/SPY)	-0.01	-0.08	-0.35	-0.98	...
Materials/S&P 500 (XLB/SPY)	-0.02	-0.07	-0.10	-0.13	-0.28
Real Estate/S&P 500 (XLRE/SPY)	-0.02	-0.09	-0.17	-0.26	-0.61
Utilities/S&P 500 (XLU/SPY)	-0.02	-0.02	-0.14	-0.41	-1.26
US/Global (SPY/ACWI)	-0.10	-0.14	-0.23	-0.63	...
Euro Area/Global (EZU/ACWI)	0.04	0.05	0.02	-0.09	-0.14
UK/Global (EWU/ACWI)	0.02	0.03	0.00	-0.06	-0.01
Canada/Global (EWC/ACWI)	-0.05	-0.05	-0.07	-0.19	-0.41
Australia/Global (EWA/ACWI)	-0.07	-0.05	-0.11	-0.23	-0.79
Japan/Global (EWJ/ACWI)	-0.09	-0.15	-0.20	-0.31	...
EM/Global (EEM/ACWI)	-0.07	-0.11	-0.24	-0.44	-0.97
China/Global (MCHI/ACWI)	0.08	0.05	0.03	-0.02	-0.37
Brazil/Global (EWZ/ACWI)	0.16	0.20	-0.02	-0.03	-0.42
India/Global (INDA/ACWI)	-0.04	-0.10	-0.03	0.00	-0.46
Long-term Bond (TLT)	-0.04	0.00	-0.04	-0.05	-0.04
Long-term Bond/S&P 500 (TLT/SPY)	-0.07	-0.16	-0.22	0.30	0.89
High-Yield Corporate (HYG)	-0.02	-0.03	-0.09	-0.23	...
High-Yield Corporate/S&P 500 (HYG/SPY)	-0.03	-0.07	-0.06	-0.41	...
DXF	0.02	0.04	-0.04	-0.08	-0.32
USD/EUR	0.01	0.04	-0.06	-0.24	...
USD/GBP	-0.01	0.00	-0.05	-0.04	-0.27
USD/CHF	0.01	0.00	-0.09	-0.14	0.18
USD/SEK	-0.01	-0.01	-0.04	-0.26	-0.16
USD/NOK	0.02	-0.02	-0.01	-0.18	-0.31
USD/CAD	-0.01	-0.04	-0.04	-0.01	...
USD/AUD	-0.06	-0.01	-0.12	-0.37	0.03
USD/NZD	-0.02	-0.05	-0.02	0.12	-0.01
USD/JPY	-0.01	0.02	0.09	0.06	-0.13
USD/CNY	0.02	0.02	0.03	0.04	0.06
USD/BRL	-0.04	-0.07	0.06	-0.02	0.13
USD/INR	-0.02	-0.03	-0.11	...	...
USD/MXN	0.00	0.01	-0.01	0.00	0.21
EUR/CHF	-0.03	-0.05	-0.04	-0.06	-0.66
EUR/GBP	-0.06	-0.12	-0.08	-0.05	...
EUR/SEK	-0.02	-0.04	-0.06	-0.16	-0.41
EUR/JPY	-0.04	-0.10	-0.28	-0.29	...
BTC	0.44	0.37	-0.07	-0.95	-3.03
ETH	0.37	0.42	0.46	-1.66	...
ETH/BTC	0.17	0.08	0.68	...	...
USO	0.06	0.12	0.65	0.92	0.58
CPER	-0.07	-0.10	-0.07	-0.06	-0.03
SLV	0.04	0.06	-0.18	0.08	0.21
GLD	0.02	-0.03	-0.20	-0.23	...
USO/CPER	0.05	0.10	0.32	0.12	-1.28
USO/GLD	0.18	0.38	0.76	0.95	1.31
CPER/GLD	-0.12	-0.12	-0.08	-0.18	-0.34
SLV/GLD	0.10	0.05	0.10	0.00	0.07

NOTE: A POSITIVE (GREEN) NET AVERAGE FORWARD RETURN MEANS THAT THE SUBSEQUENT DAY'S RETURN WAS ABOVE-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A TREND-FOLLOWING STRATEGY. IN CONTRAST, A NEGATIVE (RED) NET AVERAGE FORWARD RETURN MEANS THAT THE RETURN WAS BELOW-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A MEAN-REVERSION STRATEGY.

APPENDIX TABLE 7

Net Forward Average Return To A Trend-Following Strategy Based On The Z-Score Of Price Deviation From 200-Day Moving Average (2001-To-2025)

Z-Score Threshold:	Below 0, Above 0	Below -0.5, Above 0.5	Below -1, Above 1	Below -1.5, Above 1.5	Below -2, Above 2
S&P 500 (SPY)	-0.04	-0.02	-0.15	-0.11	-0.19
Russell 2000 (IWM)	-0.02	-0.08	-0.03	-0.03	-0.06
S&P 600 (IJR)	-0.04	-0.06	-0.03	-0.12	-0.08
Value (IVE)	-0.04	-0.03	-0.04	-0.17	-0.14
Growth (IVW)	0.00	-0.06	-0.11	-0.27	-0.43
Momentum (MTUM)	-0.03	-0.07	-0.06	-0.08	-0.21
Quality (QUAL)	-0.05	-0.06	-0.12	-0.10	-0.38
Russell 2000/S&P 500 (IWM/SPY)	-0.01	-0.03	-0.03	-0.09	-0.14
S&P 600/S&P 500 (IJR/SPY)	-0.02	-0.02	0.00	-0.04	-0.06
Value/S&P 500 (IVE/SPY)	0.00	0.00	0.00	-0.04	-0.04
Growth/S&P 500 (IVW/SPY)	0.01	0.00	0.00	-0.06	-0.08
Momentum/S&P 500 (MTUM/SPY)	0.01	0.01	-0.01	-0.01	-0.12
Quality/S&P 500 (QUAL/SPY)	0.00	0.01	0.01	0.01	0.00
Communication Services/S&P 500 (XLC/SPY)	0.06	0.08	0.06	-0.03	-0.04
Consumer Discretionary/S&P 500 (XLY/SPY)	0.00	-0.02	-0.05	-0.06	0.00
Consumer Staples/S&P 500 (XLP/SPY)	0.01	0.00	-0.15	-0.21	-0.55
Energy/S&P 500 (XLE/SPY)	0.04	0.06	0.02	0.01	0.00
Financials/S&P 500 (XLF/SPY)	-0.01	0.00	-0.02	-0.07	-0.17
Health Care/S&P 500 (XLV/SPY)	0.01	0.01	-0.05	-0.07	-0.19
Industrials/S&P 500 (XLI/SPY)	0.00	-0.03	-0.04	-0.16	-0.11
IT/S&P 500 (XLK/SPY)	-0.02	-0.07	-0.59	...	...
Materials/S&P 500 (XLB/SPY)	-0.03	-0.11	-0.18	-0.46	...
Real Estate/S&P 500 (XLRE/SPY)	-0.03	-0.05	-0.15	-0.40	-0.34
Utilities/S&P 500 (XLU/SPY)	0.00	-0.02	-0.13	-0.29	-0.64
US/Global (SPY/ACWI)	-0.04	-0.07	-0.10	-0.15	-0.19
Euro Area/Global (EZU/ACWI)	-0.05	-0.08	-0.08	-0.19	-0.31
UK/Global (EWU/ACWI)	-0.10	-0.14	-0.18	-0.25	-0.37
Canada/Global (EWC/ACWI)	0.00	-0.05	-0.11	-0.50	-0.93
Australia/Global (EWA/ACWI)	-0.11	-0.13	-0.13	-0.25	-0.58
Japan/Global (EWJ/ACWI)	-0.05	-0.08	-0.10	-0.09	-0.52
EM/Global (EEM/ACWI)	-0.04	-0.05	-0.12	-0.24	-0.68
China/Global (MCHI/ACWI)	0.03	0.00	-0.05	-0.03	-0.37
Brazil/Global (EWZ/ACWI)	0.02	0.04	-0.06	-0.47	-1.18
India/Global (INDA/ACWI)	-0.06	-0.08	-0.09	-0.21	-0.44
Long-term Bond (TLT)	0.00	-0.01	-0.05	-0.02	-0.09
Long-term Bond/S&P 500 (TLT/SPY)	-0.01	0.00	-0.04	0.00	-0.30
High-Yield Corporate (HYG)	0.01	0.01	0.03	-0.01	0.03
High-Yield Corporate/S&P 500 (HYG/SPY)	-0.07	-0.17	-0.24	-0.51	...
DXY	0.02	0.01	0.01	0.03	0.02
USD/EUR	0.02	0.02	0.01	-0.01	0.05
USD/GBP	0.01	0.01	-0.01	-0.02	0.07
USD/CHF	-0.01	-0.01	-0.05	-0.15	-0.08
USD/SEK	0.01	0.03	0.04	-0.05	0.17
USD/NOK	-0.01	-0.01	-0.01	0.03	0.12
USD/CAD	0.00	0.00	0.01	0.01	0.10
USD/AUD	0.02	0.02	0.03	0.05	0.10
USD/NZD	-0.02	-0.01	0.02	0.12	0.08
USD/JPY	0.00	0.00	-0.01	0.04	0.01
USD/CNY	0.01	0.02	0.03	0.03	0.04
USD/BRL	0.05	0.09	0.06	0.05	-0.05
USD/INR	0.01	0.03	0.04	0.05	0.01
USD/MXN	-0.01	0.00	0.02	0.01	-0.16
EUR/CHF	0.00	0.00	0.01	0.01	-0.03
EUR/GBP	-0.01	-0.02	-0.01	-0.02	-0.03
EUR/SEK	-0.01	-0.03	-0.04	-0.04	-0.03
EUR/JPY	0.00	-0.02	-0.02	0.04	0.05
BTC	0.35	0.57	0.80	...	...
ETH	0.16	0.37	0.45	...	...
ETH/BTC	0.17	0.14	0.05	-0.47	...
USO	0.05	0.26	0.46	0.35	-0.02
CPER	-0.06	-0.06	-0.07	-0.14	-0.44
SLV	0.03	0.11	-0.02	-0.03	0.20
GLD	0.01	0.01	0.01	-0.05	-0.02
USO/CPER	0.04	0.03	0.24	0.31	0.15
USO/GLD	0.13	0.18	0.33	0.34	0.26
CPER/GLD	-0.08	-0.08	-0.09	-0.14	-0.21
SLV/GLD	0.00	0.01	0.06	-0.01	0.10

NOTE: A POSITIVE (GREEN) NET AVERAGE FORWARD RETURN MEANS THAT THE SUBSEQUENT DAY'S RETURN WAS ABOVE-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A TREND-FOLLOWING STRATEGY. IN CONTRAST, A NEGATIVE (RED) NET AVERAGE FORWARD RETURN MEANS THAT THE RETURN WAS BELOW-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A MEAN-REVERSION STRATEGY.

APPENDIX TABLE 8

Net Forward Average Return To A Trend-Following Strategy Based On The Z-Score Of Price Deviation From 200-Day Moving Average (2020-To-2025)

Z-Score Threshold:	Below 0, Above 0	Below -0.5, Above 0.5	Below -1, Above 1	Below -1.5, Above 1.5	Below -2, Above 2
S&P 500 (SPY)	0.00	-0.06	-0.25	-0.41	...
Russell 2000 (IWM)	-0.02	-0.10	-0.06	-0.06	-0.25
S&P 600 (IJR)	-0.05	-0.06	0.01	-0.10	-0.29
Value (IVE)	-0.07	-0.15	-0.14	-0.51	...
Growth (IVW)	0.05	0.01	-0.16	-0.68	-1.17
Momentum (MTUM)	-0.05	-0.06	-0.07	-0.38	-0.71
Quality (QUAL)	0.04	-0.03	-0.10	-0.30	...
Russell 2000/S&P 500 (IWM/SPY)	0.02	0.00	0.09	0.01	-0.01
S&P 600/S&P 500 (IJR/SPY)	0.03	0.07	0.07	0.03	0.01
Value/S&P 500 (IVE/SPY)	0.01	0.02	0.02	-0.03	-0.04
Growth/S&P 500 (IVW/SPY)	0.02	0.01	0.02	-0.05	-0.07
Momentum/S&P 500 (MTUM/SPY)	0.00	-0.01	-0.01	-0.06	-0.30
Quality/S&P 500 (QUAL/SPY)	0.01	0.00	-0.02	-0.12	...
Communication Services/S&P 500 (XLC/SPY)	0.07	0.08	0.07	-0.02	-0.04
Consumer Discretionary/S&P 500 (XLY/SPY)	-0.03	-0.01	-0.07	-0.07	0.00
Consumer Staples/S&P 500 (XLP/SPY)	0.04	0.00	-0.16	-0.36	...
Energy/S&P 500 (XLE/SPY)	0.11	0.16	0.16	0.35	0.28
Financials/S&P 500 (XLF/SPY)	0.05	0.00	-0.03	-0.01	-0.32
Health Care/S&P 500 (XLV/SPY)	-0.01	0.04	0.00	-0.16	-0.57
Industrials/S&P 500 (XLI/SPY)	-0.01	-0.05	-0.04	-0.08	-0.07
IT/S&P 500 (XLK/SPY)	0.00	-0.05	-0.57	...	...
Materials/S&P 500 (XLB/SPY)	0.01	-0.06	-0.16	-0.36	...
Real Estate/S&P 500 (XLRE/SPY)	-0.08	-0.09	-0.29	...	...
Utilities/S&P 500 (XLU/SPY)	0.00	-0.02	-0.09	-0.23	-0.81
US/Global (SPY/ACWI)	-0.05	-0.08	-0.30	-1.20	...
Euro Area/Global (EZU/ACWI)	-0.01	-0.03	-0.03	-0.02	-0.46
UK/Global (EWU/ACWI)	-0.01	-0.01	0.01	-0.04	-0.01
Canada/Global (EWC/ACWI)	0.00	-0.06	-0.11	-0.48	...
Australia/Global (EWA/ACWI)	-0.08	-0.10	-0.03	-0.40	...
Japan/Global (EWJ/ACWI)	-0.08	-0.16	-0.23	...	...
EM/Global (EEM/ACWI)	-0.04	-0.07	-0.25	...	...
China/Global (MCHI/ACWI)	0.07	0.03	-0.05	-0.15	-0.40
Brazil/Global (EWZ/ACWI)	-0.07	-0.23	-0.32	...	...
India/Global (INDA/ACWI)	-0.03	-0.07	-0.11	-0.33	-0.59
Long-term Bond (TLT)	-0.03	0.00	0.00	-0.03	-0.10
Long-term Bond/S&P 500 (TLT/SPY)	-0.26	0.04	-0.07	0.25	...
High-Yield Corporate (HYG)	0.00	0.00	-0.06	...	...
High-Yield Corporate/S&P 500 (HYG/SPY)	0.01	-0.12	-0.19	-0.79	...
DXY	0.01	0.01	-0.03	-0.11	...
USD/EUR	0.01	0.02	-0.02	-0.08	...
USD/GBP	-0.01	-0.02	-0.02	...	...
USD/CHF	0.00	-0.03	-0.10	-0.31	...
USD/SEK	0.00	-0.01	-0.02	-0.50	...
USD/NOK	-0.02	-0.05	-0.09	-0.47	...
USD/CAD	-0.02	-0.03	-0.10	...	...
USD/AUD	-0.04	-0.08	-0.14	...	...
USD/NZD	-0.05	-0.09	-0.04	...	...
USD/JPY	-0.02	-0.07	-0.23	...	...
USD/CNY	-0.02	0.02	0.04	0.04	0.04
USD/BRL	-0.02	-0.02	-0.10	...	...
USD/INR	-0.01	-0.02	...	...	...
USD/MXN	0.01	0.03	0.02	-0.02	-0.24
EUR/CHF	-0.03	-0.04	-0.05	...	...
EUR/GBP	-0.03	-0.09	-0.47	...	...
EUR/SEK	-0.02	-0.02	-0.08	-0.28	...
EUR/JPY	-0.03	-0.16	-0.65	...	...
BTC	0.13	0.59	0.88	...	...
ETH	0.17	0.42	-0.19	...	...
ETH/BTC	0.16	0.18	0.14	-0.47	...
USO	0.05	0.41	0.58	0.26	-0.53
CPER	-0.10	-0.10	-0.02	-0.15	-0.51
SLV	-0.01	0.08	-0.19	-1.90	...
GLD	0.05	-0.02	0.00	-0.55	...
USO/CPER	0.06	0.23	0.55	0.40	0.49
USO/GLD	0.26	0.41	0.56	0.50	0.05
CPER/GLD	-0.11	-0.01	-0.01	0.06	0.04
SLV/GLD	-0.03	-0.01	-0.04	-0.17	-0.81

NOTE: A POSITIVE (GREEN) NET AVERAGE FORWARD RETURN MEANS THAT THE SUBSEQUENT DAY'S RETURN WAS ABOVE-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A TREND-FOLLOWING STRATEGY. IN CONTRAST, A NEGATIVE (RED) NET AVERAGE FORWARD RETURN MEANS THAT THE RETURN WAS BELOW-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A MEAN-REVERSION STRATEGY.

APPENDIX TABLE 9

A Simple Average Of Net Forward Returns To A Trend-Following Strategy Based On The Z-Scores Of Price Deviation From 1-Day, 10-Day, 50-Day, And 200-Day Moving Averages (2001-To-2025)

Z-Score Threshold:	Below 0, Above 0	Below -0.5, Above 0.5	Below -1, Above 1	Below -1.5, Above 1.5	Below -2, Above 2
S&P 500 (SPY)	-0.06	-0.10	-0.22	-0.33	-0.55
Russell 2000 (IWM)	-0.06	-0.11	-0.16	-0.29	-0.48
S&P 600 (IJR)	-0.06	-0.10	-0.14	-0.28	-0.45
Value (IVE)	-0.05	-0.09	-0.13	-0.29	-0.45
Growth (IVW)	-0.05	-0.12	-0.23	-0.43	-0.75
Momentum (MTUM)	-0.06	-0.10	-0.14	-0.26	-0.69
Quality (QUAL)	-0.06	-0.12	-0.21	-0.31	-0.60
Russell 2000/S&P 500 (IWM/SPY)	-0.02	-0.06	-0.08	-0.10	-0.13
S&P 600/S&P 500 (IJR/SPY)	-0.02	-0.05	-0.07	-0.10	-0.15
Value/S&P 500 (IVE/SPY)	0.01	0.01	0.00	-0.01	-0.03
Growth/S&P 500 (IVW/SPY)	0.01	0.01	0.00	-0.01	-0.08
Momentum/S&P 500 (MTUM/SPY)	0.02	0.03	0.02	-0.02	-0.07
Quality/S&P 500 (QUAL/SPY)	0.00	0.00	-0.01	-0.02	-0.03
Communication Services/S&P 500 (XLC/SPY)	-0.03	-0.03	-0.05	-0.07	-0.07
Consumer Discretionary/S&P 500 (XLY/SPY)	0.00	0.00	-0.01	-0.04	-0.07
Consumer Staples/S&P 500 (XLP/SPY)	0.02	0.00	-0.11	-0.21	-0.47
Energy/S&P 500 (XLE/SPY)	0.04	0.06	0.02	-0.02	-0.04
Financials/S&P 500 (XLF/SPY)	-0.04	-0.09	-0.20	-0.33	-0.43
Health Care/S&P 500 (XLV/SPY)	0.02	0.02	0.00	-0.08	-0.23
Industrials/S&P 500 (XLI/SPY)	0.01	-0.01	-0.03	-0.06	-0.15
IT/S&P 500 (XLK/SPY)	-0.02	-0.07	-0.30	-0.51	-0.41
Materials/S&P 500 (XLB/SPY)	-0.02	-0.06	-0.13	-0.21	-0.17
Real Estate/S&P 500 (XLRE/SPY)	-0.09	-0.17	-0.37	-0.89	-1.46
Utilities/S&P 500 (XLU/SPY)	-0.01	-0.02	-0.11	-0.28	-0.52
US/Global (SPY/ACWI)	-0.14	-0.26	-0.43	-0.68	-0.88
Euro Area/Global (EZU/ACWI)	-0.16	-0.26	-0.39	-0.55	-0.80
UK/Global (EWU/ACWI)	-0.17	-0.28	-0.47	-0.70	-0.97
Canada/Global (EWC/ACWI)	-0.11	-0.20	-0.33	-0.71	-1.06
Australia/Global (EWA/ACWI)	-0.18	-0.32	-0.53	-0.86	-1.48
Japan/Global (EWJ/ACWI)	-0.07	-0.11	-0.19	-0.30	-0.73
EM/Global (EEM/ACWI)	-0.16	-0.28	-0.52	-0.97	-1.61
China/Global (MCHI/ACWI)	0.00	-0.03	-0.06	-0.08	-0.24
Brazil/Global (EWZ/ACWI)	-0.05	-0.07	-0.25	-0.61	-1.16
India/Global (INDA/ACWI)	-0.06	-0.08	-0.14	-0.29	-0.69
Long-term Bond (TLT)	-0.03	-0.04	-0.08	-0.10	-0.21
Long-term Bond/S&P 500 (TLT/SPY)	-0.08	-0.13	-0.32	-0.41	-0.72
High-Yield Corporate (HYG)	0.02	0.03	0.06	0.09	-0.02
High-Yield Corporate/S&P 500 (HYG/SPY)	-0.09	-0.20	-0.34	-0.69	-1.66
DXE	0.00	0.01	0.00	0.00	-0.03
USD/EUR	0.00	0.01	0.02	-0.01	0.03
USD/GBP	0.00	-0.01	-0.01	0.04	0.07
USD/CHF	-0.01	-0.01	-0.03	-0.03	0.01
USD/SEK	-0.01	0.00	-0.01	-0.06	-0.05
USD/NOK	-0.01	0.00	-0.02	-0.02	0.01
USD/CAD	-0.01	-0.01	-0.01	0.04	0.09
USD/AUD	0.01	0.01	-0.01	0.00	-0.03
USD/NZD	0.00	0.00	0.02	0.02	-0.02
USD/JPY	-0.01	0.00	-0.02	-0.02	-0.09
USD/CNY	0.00	0.00	0.00	0.01	0.01
USD/BRL	0.04	0.05	0.05	0.02	-0.02
USD/INR	0.01	0.02	0.03	0.02	0.00
USD/MXN	0.01	0.01	0.00	0.01	0.01
EUR/CHF	-0.01	0.00	0.00	0.00	-0.02
EUR/GBP	-0.02	-0.02	-0.03	-0.05	-0.05
EUR/SEK	-0.02	-0.03	-0.05	-0.07	-0.09
EUR/JPY	-0.01	-0.01	-0.02	0.00	-0.03
BTC	0.40	0.51	0.60	-0.45	-1.34
ETH	0.18	0.26	0.03	-1.85	-4.00
ETH/BTC	0.16	0.24	-0.08	-0.77	0.63
USO	0.05	0.14	0.28	0.23	-0.03
CPER	-0.09	-0.13	-0.14	-0.16	-0.37
SLV	0.05	0.06	0.00	-0.03	0.18
GLD	0.00	0.00	-0.04	-0.06	-0.21
USO/CPER	0.00	0.01	0.13	0.05	-0.07
USO/GLD	0.12	0.16	0.29	0.27	0.11
CPER/GLD	-0.11	-0.16	-0.20	-0.26	-0.38
SLV/GLD	0.02	0.02	0.08	0.08	0.08

NOTE: A POSITIVE (GREEN) NET AVERAGE FORWARD RETURN MEANS THAT THE SUBSEQUENT DAY'S RETURN WAS ABOVE-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A TREND-FOLLOWING STRATEGY. IN CONTRAST, A NEGATIVE (RED) NET AVERAGE FORWARD RETURN MEANS THAT THE RETURN WAS BELOW-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A MEAN-REVERSION STRATEGY.

APPENDIX TABLE 10

A Simple Average Of Net Forward Returns To A Trend-Following Strategy Based On The Z-Scores Of Price Deviation From 1-Day, 10-Day, 50-Day, And 200-Day Moving Averages (2020-To-2025)

Z-Score Threshold:	Below 0, Above 0	Below -0.5, Above 0.5	Below -1, Above 1	Below -1.5, Above 1.5	Below -2, Above 2
S&P 500 (SPY)	-0.05	-0.10	-0.26	-0.46	-0.88
Russell 2000 (IWM)	-0.04	-0.08	-0.14	-0.24	-0.48
S&P 600 (IJR)	-0.08	-0.09	-0.13	-0.21	-0.40
Value (IVE)	-0.04	-0.08	-0.11	-0.26	-0.74
Growth (IVW)	-0.05	-0.08	-0.23	-0.51	-0.80
Momentum (MTUM)	-0.08	-0.06	-0.19	-0.33	-0.80
Quality (QUAL)	-0.04	-0.09	-0.17	-0.37	-0.72
Russell 2000/S&P 500 (IWM/SPY)	-0.01	-0.04	0.02	0.02	0.07
S&P 600/S&P 500 (IJR/SPY)	0.00	-0.03	-0.01	0.02	-0.02
Value/S&P 500 (IVE/SPY)	0.02	0.02	0.01	0.01	-0.02
Growth/S&P 500 (IVW/SPY)	0.02	0.02	0.01	0.01	-0.07
Momentum/S&P 500 (MTUM/SPY)	0.01	0.01	0.02	0.00	-0.10
Quality/S&P 500 (QUAL/SPY)	0.00	-0.01	-0.02	-0.05	-0.03
Communication Services/S&P 500 (XLC/SPY)	-0.02	-0.03	-0.05	-0.06	-0.06
Consumer Discretionary/S&P 500 (XLY/SPY)	-0.01	0.00	-0.01	-0.06	-0.12
Consumer Staples/S&P 500 (XLP/SPY)	0.00	-0.01	-0.08	-0.18	-0.36
Energy/S&P 500 (XLE/SPY)	0.15	0.21	0.22	0.29	0.24
Financials/S&P 500 (XLF/SPY)	0.00	-0.01	-0.04	-0.10	-0.27
Health Care/S&P 500 (XLV/SPY)	0.00	0.02	0.00	-0.08	-0.29
Industrials/S&P 500 (XLI/SPY)	0.01	-0.02	-0.04	-0.03	-0.13
IT/S&P 500 (XLK/SPY)	-0.02	-0.07	-0.29	-0.56	-0.76
Materials/S&P 500 (XLB/SPY)	0.00	-0.05	-0.05	-0.09	-0.03
Real Estate/S&P 500 (XLRE/SPY)	-0.07	-0.11	-0.16	-0.26	-0.52
Utilities/S&P 500 (XLU/SPY)	-0.05	-0.05	-0.14	-0.26	-0.71
US/Global (SPY/ACWI)	-0.13	-0.23	-0.44	-0.92	-1.62
Euro Area/Global (EZU/ACWI)	-0.04	-0.07	-0.08	0.03	0.02
UK/Global (EWU/ACWI)	-0.04	-0.04	-0.08	-0.14	-0.15
Canada/Global (EWC/ACWI)	-0.04	-0.08	-0.11	-0.33	-0.63
Australia/Global (EWA/ACWI)	-0.16	-0.27	-0.44	-0.99	-1.91
Japan/Global (EWJ/ACWI)	-0.06	-0.08	-0.15	-0.23	-0.12
EM/Global (EEM/ACWI)	-0.08	-0.16	-0.28	-0.48	-1.16
China/Global (MCHI/ACWI)	0.02	0.00	-0.04	-0.14	-0.37
Brazil/Global (EWZ/ACWI)	-0.04	-0.07	-0.22	-0.50	-1.05
India/Global (INDA/ACWI)	-0.05	-0.09	-0.21	-0.67	-1.40
Long-term Bond (TLT)	-0.03	-0.03	-0.09	-0.18	-0.26
Long-term Bond/S&P 500 (TLT/SPY)	-0.18	-0.17	-0.42	-0.47	-0.92
High-Yield Corporate (HYG)	-0.02	-0.01	0.00	-0.03	-0.04
High-Yield Corporate/S&P 500 (HYG/SPY)	-0.04	-0.11	-0.22	-0.55	-1.22
DXY	0.02	0.02	-0.02	-0.05	-0.10
USD/EUR	0.00	0.01	-0.04	-0.06	-0.03
USD/GBP	-0.02	-0.01	-0.05	-0.02	-0.11
USD/CHF	0.01	0.01	-0.02	-0.09	0.19
USD/SEK	-0.01	-0.03	-0.05	-0.24	-0.13
USD/NOK	-0.02	-0.03	-0.02	-0.10	0.07
USD/CAD	-0.02	-0.03	-0.06	0.07	0.15
USD/AUD	-0.05	-0.07	-0.11	-0.10	0.13
USD/NZD	-0.05	-0.07	-0.04	0.06	0.21
USD/JPY	-0.01	-0.01	-0.06	-0.04	-0.15
USD/CNY	0.00	0.00	0.00	0.00	0.00
USD/BRL	0.00	-0.01	-0.02	-0.21	-0.26
USD/INR	-0.01	-0.03	-0.09	-0.12	-0.33
USD/MXN	0.00	0.01	0.00	-0.01	0.04
EUR/CHF	-0.02	-0.03	-0.05	-0.13	-0.44
EUR/GBP	-0.04	-0.08	-0.18	-0.09	0.15
EUR/SEK	-0.03	-0.05	-0.09	-0.21	-0.33
EUR/JPY	-0.03	-0.10	-0.29	-0.23	-0.21
BTC	0.16	0.28	0.16	-1.36	-2.01
ETH	0.14	0.24	-0.30	-2.20	-4.51
ETH/BTC	0.13	0.20	0.02	-0.94	0.23
USO	0.09	0.26	0.63	0.77	0.86
CPER	-0.09	-0.11	-0.07	-0.08	-0.20
SLV	0.01	0.07	-0.03	-0.44	0.61
GLD	0.03	-0.01	-0.07	-0.16	0.16
USO/CPER	0.04	0.11	0.41	0.50	0.15
USO/GLD	0.22	0.33	0.63	0.82	0.65
CPER/GLD	-0.15	-0.16	-0.15	-0.13	-0.23
SLV/GLD	0.05	0.06	0.08	0.05	-0.04

NOTE: A POSITIVE (GREEN) NET AVERAGE FORWARD RETURN MEANS THAT THE SUBSEQUENT DAY'S RETURN WAS ABOVE-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A TREND-FOLLOWING STRATEGY. IN CONTRAST, A NEGATIVE (RED) NET AVERAGE FORWARD RETURN MEANS THAT THE RETURN WAS BELOW-AVERAGE WHEN THE PRIOR DAY'S Z-SCORE WAS POSITIVE, SIGNIFYING SUPERIOR RETURNS TO A MEAN-REVERSION STRATEGY.

Global Investment Strategy View Matrix

The Global Investment Strategy team (GIS) summarizes its views via a matrix featuring recommendations over a 12 month horizon. While GIS uses the MacroQuant model as one of several inputs into its decision-making process, the GIS views shown in this matrix may differ from the MacroQuant model's recommendations shown in the dedicated monthly MacroQuant reports.

Global Investment Strategy Views (12 Month)				
Global Asset Allocation	-	0	+	
Equities		■		
Fixed Income			■	
Cash				■
Global Equities (Region)*	-	0	+	
US			■	
Euro Area		■		
UK		■		
Canada			■	
Australia			■	
Japan			■	
EM				■
Global Equities (Sectors)*	-	0	+	
Communication Services			■	
Consumer Discretionary		■		
Consumer Staples				■
Energy			■	
Financials		■		
Health Care				■
Industrials		■		
Information Technology			■	
Materials			■	
Real Estate		■		
Utilities				■
Global Fixed Income**	-	0	+	
Government				■
Investment Grade		■		
High Yield		■		
Duration				■
Inflation Protection			■	
Currencies***	-	0	+	
USD		■		
EUR			■	
GBP		■		
CHF				■
JPY				■
CAD		■		
AUD			■	
CNY				■
Commodities (vs USD)	-	0	+	
Energy		■		
Base/Bulk Metals			■	
Gold				■

LEGEND:

■ - CURRENT
□ - PREVIOUS

1 (MOST LEFT) - STRONG SELL

2 - SELL

3 - NEUTRAL

4 - BUY

5 (MOST RIGHT) - STRONG BUY

* RELATIVE TO MSCI ACWI (CURRENCY UNHEDGED).

** RELATIVE TO BLOOMBERG GLOBAL AGGREGATE (CURRENCY-HEDGED).

***IN TRADE-WEIGHTED TERMS.

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